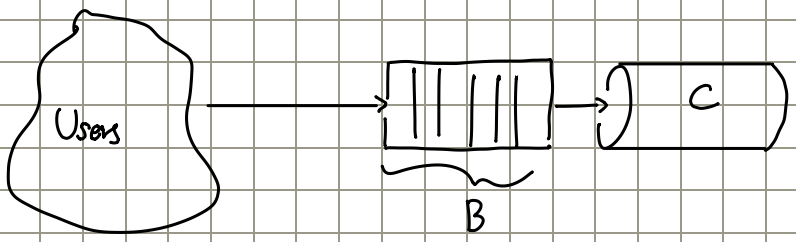
Markov Inequality

$\forall x \geq 0 \quad \forall a \in \mathbb{R}^+ \quad P[x \geq a] \leq \frac{E[x]}{a}$

Chebishev

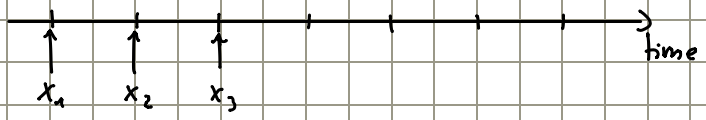
$P[|x - m_x| > a] \leq \frac{\sigma_x^2}{a^2}$

Problem



X_i = # of bytes generated at time interval δ_i = const =

$\{X_i\}$ are i.i.d $\quad m_x \triangleq E[X_i] \quad \sigma_x^2 = E[(X_i - m_x)]^2$



In each slot the link can transmit at most $\Delta = C \cdot \delta$ bytes

$$P[X > a] \leq \frac{m_x}{a} \; ; \quad P[|x - m_x| > a] \leq \frac{\sigma_x^2}{a^2}$$

Find C and B

$C = \frac{\Delta}{\tau}$ → bytes transmitted in each slot

↳ slot time

Supposition: no buffer

Congestion prob. $\leq \varepsilon = 10^{-2}$

Congestion if $x_i > \Delta$. We want to limit $\underbrace{P[x_i > \Delta]}_{P[x_i > \Delta] \leq \frac{m_x}{\Delta} < \varepsilon} \leq \varepsilon$

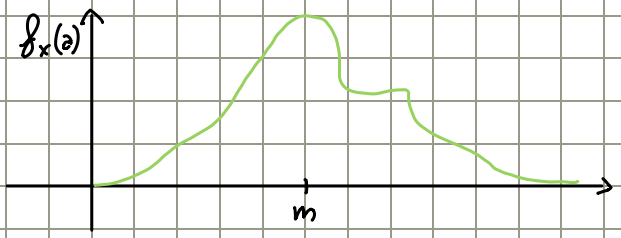
$\frac{m_x}{\Delta} \overset{!}{\leq} \varepsilon$ → this is a condition that we set

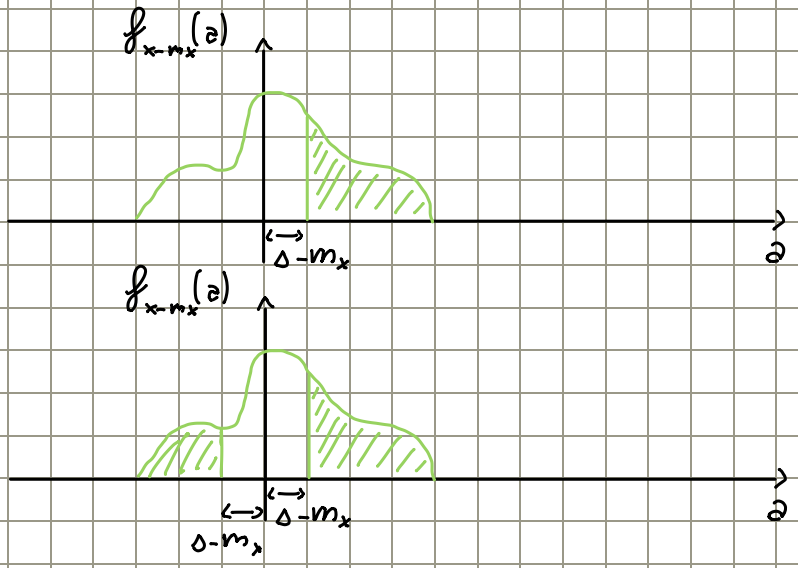
$\frac{m_x}{c \cdot \delta} \leq \varepsilon \Rightarrow c \cdot \delta \geq \frac{m}{\varepsilon} \Rightarrow c \geq \frac{m}{\delta \varepsilon} = 100 \cdot \frac{m}{\delta}$ ($\varepsilon = 10^{-2}$)

Now we try to use the Chebyshev bound

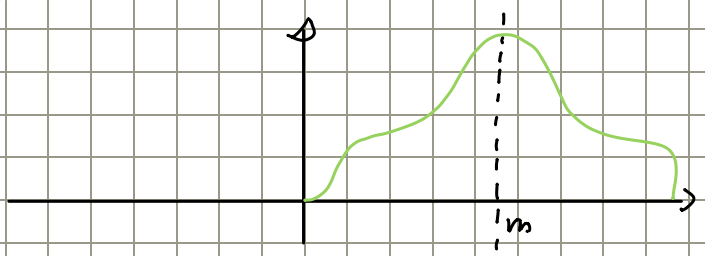
$P[x > \Delta] = P[x - m_x > \Delta - m_x]$ ($\Delta > m_x$)

$\leq \underbrace{P[|x - m_x| > \Delta - m_x]}_{P[x - m_x > \Delta - m_x \cup m_x - x > \Delta - m_x]} \leq$ Cheb. $\frac{\delta^2}{(\Delta - m_x)^2} \overset{!}{\leq} \varepsilon \Rightarrow$





If $f(x)$ is symmetric w.r.t. the mean of x:



then $P[|x - m_x| > a] = 2P[x - m_x > a]$

So, applying Cheb. bound:

$$P[|x-m_x| > \Delta - m_x] \overset{\text{Cheb.}}{\leq} \frac{\sigma^2}{(\Delta-m_x)^2} \overset{!}{\leq} \varepsilon \Rightarrow \frac{\sigma_x^2}{\varepsilon} \leq (\Delta - m_x)^2$$

$$\frac{\sigma_x}{\sqrt{\varepsilon}} \leq \Delta - m_x$$

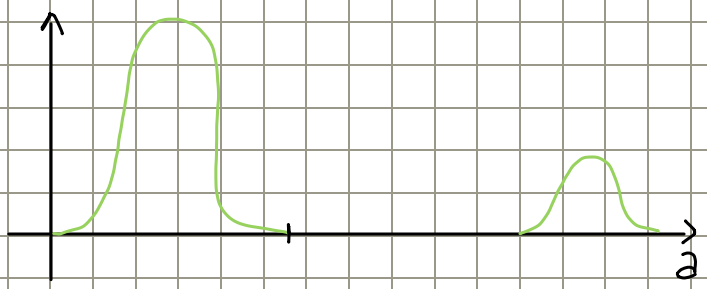
$$\frac{\sigma_x}{\sqrt{\varepsilon}} \leq C\delta - m_x$$

$$\Rightarrow C \geq \frac{\sigma_x}{\delta\sqrt{\varepsilon}} + \frac{m_x}{\delta}$$

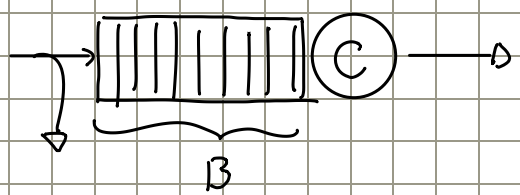
Markov: $C > \frac{m}{\delta} \cdot \frac{1}{\varepsilon}$

Chebishev: $C > \frac{m}{\delta} + \frac{\delta_x}{\delta\sqrt{\varepsilon}} = \frac{m}{\delta} + \frac{\sigma_x}{\delta} \cdot 10$ ($\varepsilon = 10^{-2}$)

If $\sigma_x > 9{,}9\, m_x \rightarrow C_m < C_c$ but this condition is very unlikely



This is fine if we don't have a buffer... but if we have a buffer?
What is the congestion condition change if we have a buffer?

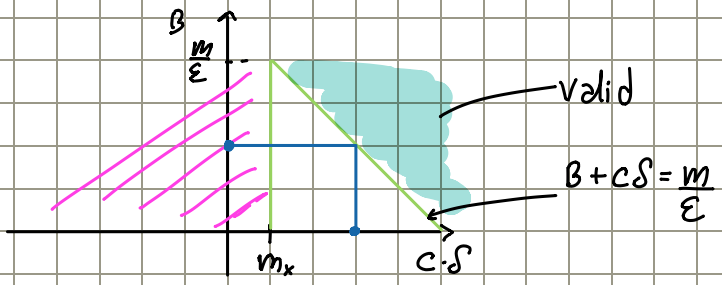


q_i = buffer level at the beginning of i-th slot

$$P[q_i + x_i > B + \Delta] = P[x_i > B + \Delta - q_i]$$

Suppose $q_i = 0$

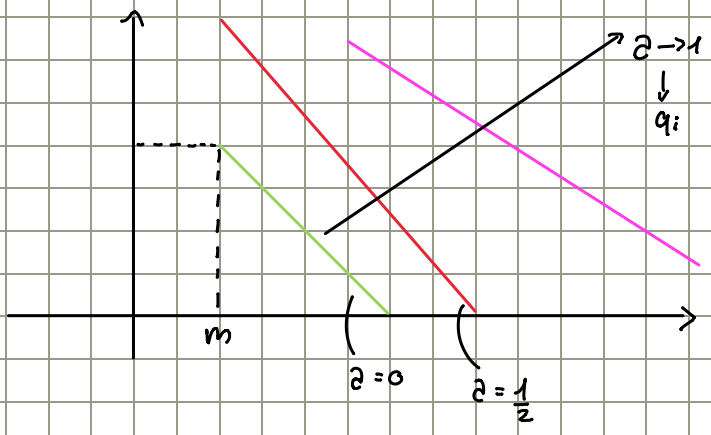
$$\rightarrow P[x_i > B + \Delta] \overset{\text{Markov}}{\leq} \frac{m_x}{B+\Delta} \overset{!}{\leq} \varepsilon \rightarrow \begin{cases} B + c\delta \geq \frac{m_x}{\varepsilon} \\ c\delta > m_x \end{cases}$$



Suppose $q_i = \alpha B \qquad \alpha \in [0,1]$

$$P[x_i > B + \Delta - \alpha B] \overset{\text{MK}}{\leq} \frac{m_x}{(1-\alpha)B + \Delta} \leq \varepsilon \Rightarrow (1-\varepsilon)B + c\delta \geq \frac{m_x}{\varepsilon}$$

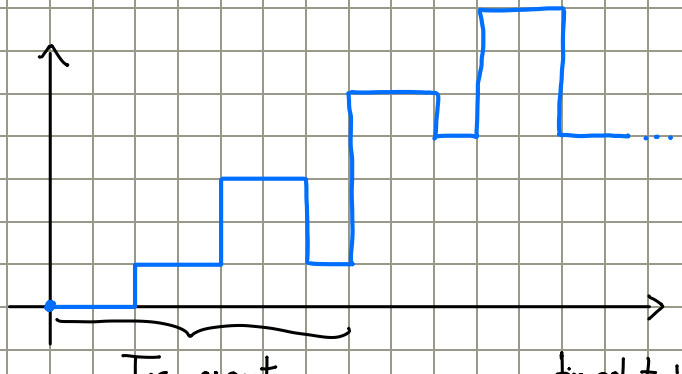
$$B \geq \left(\frac{m_x}{\varepsilon} - c\delta\right) \cdot \frac{1}{1-\alpha}$$



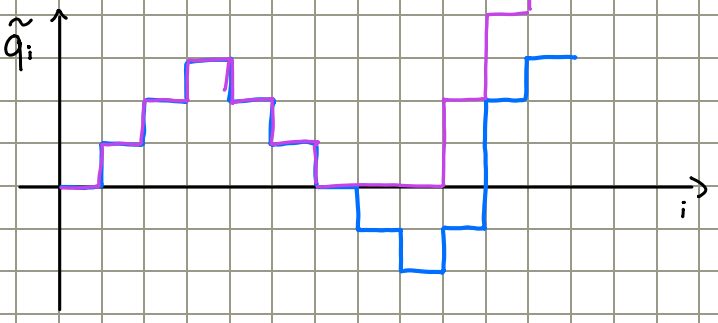
$$q_{i+1} = \max[q_i + x_i - \Delta, 0] \geq \underbrace{q_i + x_i - \Delta}_{\triangleq \tilde{q}_{i+1}}$$

$q_0 = 0, \quad q_1 = \max(x_0 - \Delta, 0)$

$q_2 = \max(q_1 + x_1 - \Delta, 0) \ldots$



$$\tilde{q}_{i+1} \overset{!}{=} \tilde{q}_i + x_i - \Delta$$



$$\tilde{q}_i \leq q_i$$

$$E[\tilde{q}_{i+1}] = E[\tilde{q}_i + x_i - \Delta]$$

$$m_q = \underbrace{E[\tilde{q}_i]}_{m_{iq}} + \underbrace{E[x_i]}_{m_x} - \underbrace{E[\Delta]}_{\Delta}$$

$$E[(\tilde{q}_{i+1})^2] = E[(q_i^2 + x_i - \Delta)^2]$$

$$\downarrow$$

$$M_q = E[\tilde{q}_i^2 + x_i^2 + \Delta^2 + 2q_i^2 x_i - 2\tilde{q}_i\Delta - 2x_i\Delta]$$

$$= M_q + \underbrace{M_x}_{m_x^2 + \sigma_x^2} + \Delta^2 + 2\underbrace{E[\tilde{q}_i^2 x_i]}_{m_q \cdot m_x} - 2\Delta m_q - 2\Delta m_x$$

$$m_{\tilde{q}} = \frac{m_x^2 + \sigma_x^2 + \Delta^2 - 2\Delta m_x}{2\Delta - 2m_x} \leq m_q$$

